Table 1: Skill routing performance using Qwen3.5 4B. BM25, Vector, and SkillSight are shown first as native retrieval baselines. Each remaining row is the equal-weight mean across those three retrievers. Columns report nDCG@1 (N@1) and nDCG@10 (N@10) for each benchmark; Average uses each retriever's task-weighted scores over 748 tasks. Bold and underlined values mark the best and second-best distinct scores in each column.

Method	BigCode Bench		CHAMP		LogicBench		MedCalc-Bench		TheoremQA		ToolQA		Average	
	N@1	N@10	N@1	N@10	N@1	N@10	N@1	N@10	N@1	N@10	N@1	N@10	N@1	N@10
Retrievers														
BM25	0.610	0.544	0.423	0.410	0.158	0.220	0.315	0.462	0.524	0.659	0.070	0.271	0.302	0.410
Vector	0.504	0.407	0.346	0.463	0.105	0.248	**0.988**	**0.992**	0.695	0.833	0.312	0.413	0.516	0.575
SkillSight	0.724	0.616	0.308	0.429	0.167	0.406	0.830	0.903	0.733	0.852	0.102	0.390	0.471	0.609
Routers														
LLM Select	0.480	0.510	0.385	0.458	0.266	0.394	0.745	0.834	0.683	0.819	0.437	0.547	0.519	0.616
SkillRouter	0.366	0.299	0.449	0.476	0.269	0.416	0.974	0.981	**0.816**	**0.860**	0.600	0.711	0.619	0.671
SkillDAG	0.612	0.600	0.359	0.453	0.143	0.378	0.709	0.785	0.651	0.781	0.161	0.421	0.429	0.576
Skill-Pro	0.238	0.446	0.218	0.395	0.161	0.317	0.705	0.850	0.460	0.712	0.402	0.527	0.407	0.574
$\text{RouterBSI}_{\text{init}}$	0.466	0.417	0.346	0.435	0.184	0.341	0.830	0.877	0.559	0.726	0.290	0.453	0.462	0.561
RouterBSI	**0.832**	**0.707**	**0.551**	**0.556**	**0.336**	**0.449**	0.915	0.933	0.686	0.808	**0.631**	**0.712**	**0.687**	**0.728**